Assignment:

1. Z-test (Two-Tailed)

A juice company claims that the average volume of its bottles is 1000 ml. A random sample of 49 bottles has a sample mean of 995 ml. The population standard deviation is 14 ml. Test the claim at the 5% significance level.

→ Given:

$\mu = 1000 ml$

$\bar{x} = 995 ml$

$\sigma = 14 ml$

$n = 49$

$\alpha = 0.05$

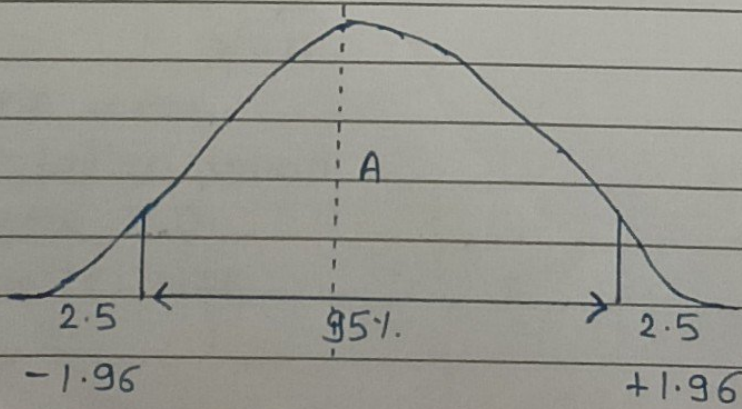
$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{995 - 1000}{14 / \sqrt{49}}$$

$$= \frac{-5}{14/7}$$

$$= \frac{-5}{2}$$

$$= -2.5$$

∴ Critical value : ± 1.96



Rejected

2. Z-Test (Right-Tailed)

A cement company claims that the average weight of a cement bag is 50 kg. A sample of 64 bags has a mean weight of 50.8 kg. The population standard deviation is 2.4 kg. Test whether the average weight is greater than 50 kg at the 5% significance level.

→ Given:

$\mu = 50\,kg$

$\bar{x} = 50.8\,kg$

$\sigma = 2.4\,kg$

$n = 4$

$\alpha = 0.05$

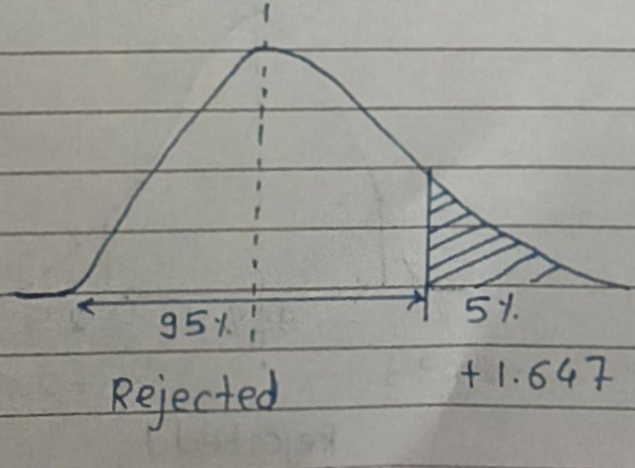
$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{50.8 - 50}{2.4/\sqrt{64}}$$

$$= \frac{0.8}{2.4/8}$$

$$= \frac{0.8}{0.3}$$

$$= 2.67$$

∴ critical value : $+1.647$



3. t-Test (Two-Tailed)

A university claims that the average statistics marks are 70. A random sample of 20 students has a mean of 73 marks and a sample standard deviation of 8 marks. Test the claim at the 5% significance level.

→ Given:

$\mu = 70$

$\bar{x} = 73$

$S = 8$

$n = 20$

$\alpha = 0.05$

$$T = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{73-70}{8/\sqrt{20}}$$

$$= \frac{3}{8/4.47}$$

$$= \frac{3}{1.7}$$

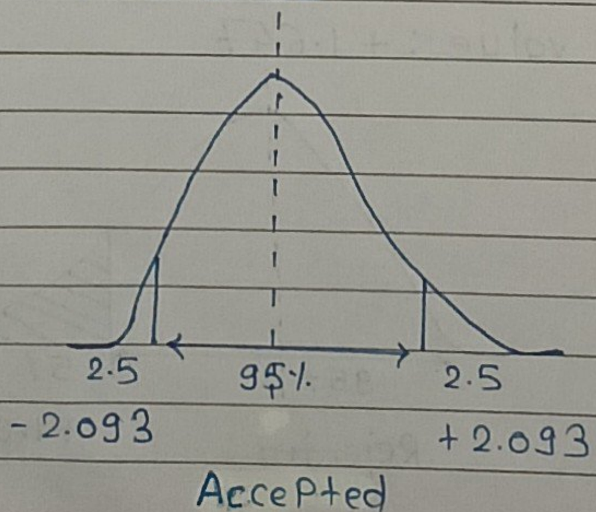
$$t = 1.76$$

$dF = n-1$

$= 20-1$

$= \underline{19}$

∴ Critical value: $\pm\ 2.093$



Accepted